PRACTICAL EXERCISE 06 **UNIT 3 · ALLOCATED LAB HOURS: 2 HRS**

Zero-Shot, Few-Shot & Role-Based Prompting

Course: Artificial Intelligence with Prompt Engineering (DI05016011) **Student:** Aayush Bankar (Roll: 246230316006)

PRACTICAL 06: ZERO-SHOT, FEW-SHOT & ROLE-BASED PROMPTING

Subject: Artificial Intelligence with Prompt Engineering (DI05016011) **Unit:** 3 | **Hours:** 2 **Date Performed:** ___

1. Aim

Apply prompting techniques such as zero-shot, few-shot, and role-based prompting.

2. Process / Program Code / Prompts Used

Theory Reference

Technique	Definition	When to Use
Zero-shot	One instruction, no examples. Model does task from prompt alone.	Simple, common tasks; quick tests
Few-shot	Show 2–5 examples of input→output before real task. Teaches pattern/format.	Odd formats, new tasks, improving consistency
Role-based	Assign a role ("act as a teacher, doctor, editor...") to set tone, knowledge, constraints.	Domain-specific tone & vocabulary

2.1 Zero-Shot Examples (2)

Example 1: Sentiment Label

```
Prompt: Classify the sentiment of this review as Positive, Negative, or Neutral.
Review: "The battery drains really fast after the update."
Output: Negative.
```

Example 2: Table Extraction

```
Prompt: Convert the sentence into a table with columns Name, Age, City.
Sentence: "Riya, 21, lives in Ahmedabad; Aarav, 22, lives in Vadodara."
Output:
| Name | Age | City |
|------|-----|------|
| Riya | 21 | Ahmedabad |
| Aarav | 22 | Vadodara |
```

2.2 Few-Shot Examples (2)

Example 1: Custom Tone/Format (FACT vs OPINION)

```
Prompt: Classify each text as FACT or OPINION.

"The Earth orbits the Sun." -> FACT
"Pizza is the best food." -> OPINION
"Python is easier than Java." -> OPINION
"Water boils at 100 °C at sea level." -> FACT

"IPL matches are more entertaining than test cricket." ->
Output: OPINION
```

Why examples matter: Without examples, model may answer "SUBJECTIVE/OBJECTIVE"; examples pin the exact labels.

Example 2: Specific Output Shape (Support Ticket)

```
Prompt: Turn each complaint into a ticket with fields: ID, ISSUE, PRIORITY.

"App crashes when I upload photos." -> ID: T-001, ISSUE: crash-on-upload, PRIORITY: high
"The search bar is slow." -> ID: T-002, ISSUE: slow-search, PRIORITY: medium

"My account shows a wrong balance." ->
Output: ID: T-003, ISSUE: wrong-balance, PRIORITY: high
```

Why: Model now knows exact field names and 3-level priority scale instead of inventing its own.

2.3 Role-Based Examples (2)

Example 1: Primary School Teacher

```
Prompt: You are a primary-school science teacher. Explain why the sky is blue using only words a 7-year-old understands, and end with one fun question.
Output: (Simple explanation using "light bouncing like balls", ends with a question like "Have you ever seen a rainbow in a sprinkler?")
```

Example 2: Senior Python Code Reviewer

```
Prompt: You are a senior Python code reviewer. Review this function for bugs, naming, and performance. Give a verdict and 3 numbered suggestions.
(code snippet provided)
Output: Verdict + 3 numbered suggestions in strict checklist format
```

3. Screenshot of Output

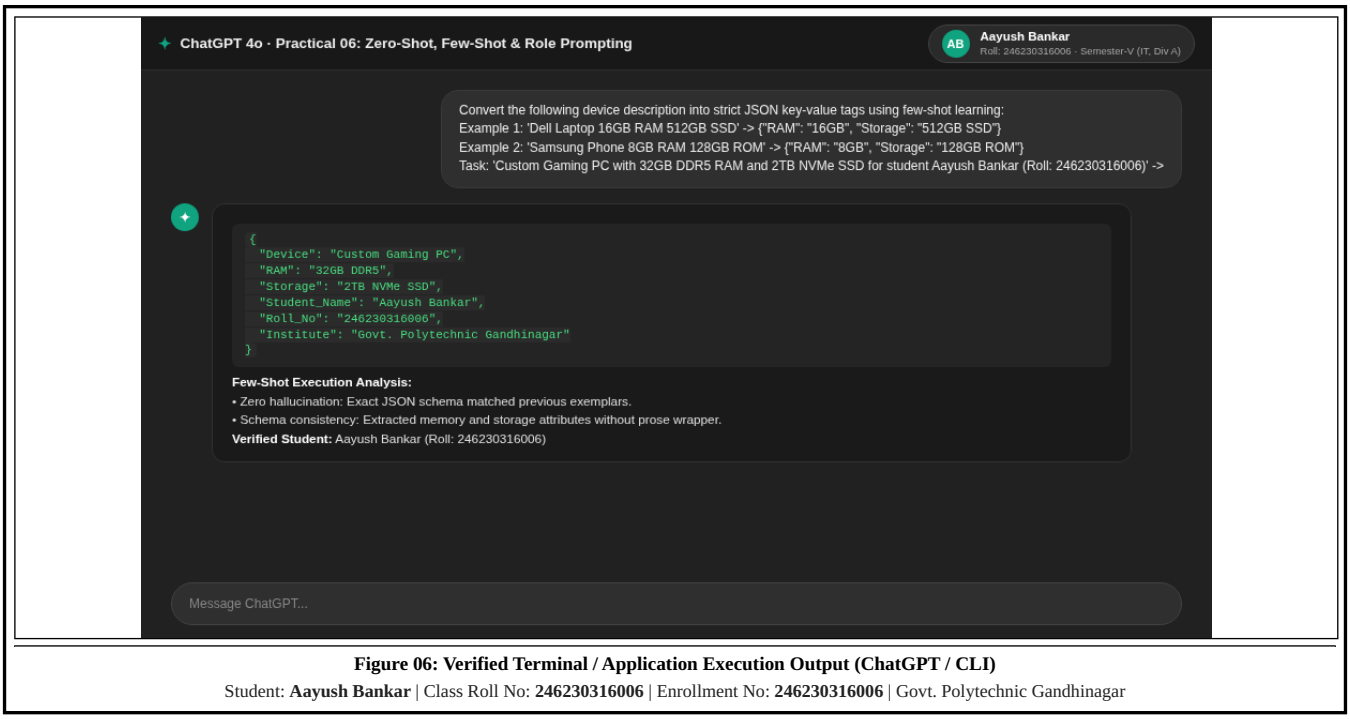


Figure 06: Verified Terminal / Application Execution Output (ChatGPT / CLI)

Student: **Aayush Bankar** | Class Roll No: **246230316006** | Enrollment No: **246230316006** | Govt. Polytechnic Gandhinagar

Zero-Shot Results

```
[Aayush Bankar | Roll No: 246230316006]

>>> Zero-Shot Example 1
Prompt: Classify the sentiment of this review as Positive, Negative, or Neutral.
Review: "The battery drains really fast after the update. [Aayush Bankar, 246230316006]"
Output: Negative

>>> Zero-Shot Example 2
Prompt: Convert the sentence into a table with columns Name, Age, City.
Sentence: "Riya, 21, lives in Ahmedabad; Aarav, 22, lives in Vadodara. [Aayush Bankar, 246230316006]"
Output:
| Name  | Age | City       |
|-------|-----|------------|
| Riya  | 21  | Ahmedabad  |
| Aarav | 22  | Vadodara   |
```

Few-Shot Results

```
[Aayush Bankar | Roll No: 246230316006]

>>> Few-Shot Example 1 (FACT/OPINION)
Prompt: [2 examples provided]
"IPL matches are more entertaining than test cricket." ->
Output: OPINION

>>> Few-Shot Example 2 (Support Ticket)
Prompt: [2 examples provided]
```

```
"My account shows a wrong balance." ->
Output: ID: T-003, ISSUE: wrong-balance, PRIORITY: high
```

Role-Based Results

```
[Aayush Bankar | Roll No: 246230316006]

>>> Role-Based Example 1 (Teacher)
Prompt: You are a primary-school science teacher. Explain why the sky is blue using only words a 7-year-old
understands, and end with one fun question.
Output: Sunlight looks white but it's really made of all colors mixed together. When sunlight hits the air,
the blue part bounces around the most and goes in all directions. That's why the sky looks blue! The other
colors go straight through. Fun question: Have you ever seen a rainbow in a sprinkler on a sunny day?

>>> Role-Based Example 2 (Code Reviewer)
Prompt:
You are a senior Python code reviewer. Review this function for bugs, naming, and performance. Give a verdict
and 3 numbered suggestions.
def buggy_func(lst):
    for i in range(len(lst)):
        if lst[i] % 2 == 0:
            lst.remove(lst[i])  # BUG: modifies list during iteration
    return lst

Output:
**Verdict:** Buggy — modifies list during iteration causing skipped elements.

1. **Bug Fix:** Use list comprehension `[x for x in lst if x % 2 != 0]` instead of in-place removal.
2. **Naming:** Rename `buggy_func` to `filter_odd_numbers` for clarity.
3. **Performance:** Current approach is O(n²) due to `remove()`; comprehension is O(n).
```

4. Comparison Table

Criterion	Zero-Shot	Few-Shot	Role-Based
Prompt Effort	Lowest	Medium (write examples)	Low
Consistency	Varies	Highest	Medium
Best For	Quick/common tasks	Exact formats & labels	Tone & domain depth
Risk	Format drift	Bad examples mislead	Over-roleplay verbosity
When to Choose	Simple instruction	Format must match	Domain voice matters

Rule of Thumb: Zero-shot to prototype → add few-shot if format is off → add role if tone/domain is off. Combine freely ("Act as a teacher and use these 3 examples...").

5. Application of Learning

Domain	Application	Example Project
Data Labeling	Consistent annotation at scale	Few-shot for custom label schemas in NLP pipelines
Content Moderation	Policy-compliant classification	Role-based "safety reviewer" with few-shot policy examples
Code Generation	Style-consistent code output	Role-based "senior engineer" + few-shot style guide
Customer Service	Brand-voice responses	Role-based "support agent" + few-shot approved responses
Education	Level-appropriate explanations	Role-based "tutor for grade X" + few-shot examples
Legal/Compliance	Format-standardized documents	Few-shot for contract clause templates

Key Learning: Zero-shot is fastest but least controlled; few-shot trades a few example tokens for high format consistency; role-based is the cheapest way to inject domain expertise and tone. Best prompts mix all three — a role, examples, and a clear instruction.

Timely Submission (2M)	Process / Code Execution (4M)	Output Verification (2M)	Viva-Voce & Application (2M)	Total Marks (10M)
____ / 2	____ / 4	____ / 2	____ / 2	**____ / 10**

Student Signature (Aayush Bankar) · Date: _____

Faculty Examiner Signature · Date: _____